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VectorScopeDefTopic

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Vector Scope

[Vector Scope](#)

The Vector Scope is a graphical tool that plots complex waveform data, in either polar or rectangular coordinates. Although the Vector Scope can plot any complex data set, it is most useful for plotting modulated and demodulated RF signals captured by the MW Receiver during test program debugging.

The Vector Scope includes the following features:

- Integration with TDE for program debugging using the VBT DSPWave.Plot method, including data for multiple sites
- Operation as a stand-alone tool for viewing files off-line
- Displays data files stored in either IG-XL DSPWave (.txt) or IMAGE Wave (.wav) formats
- Loads more than one data set (10 maximum) and allows control of data shown
- Saves data sets and plots, including modifications, in .txt format files
- Interactive analysis of data points and data sets using zoom, markers, error vector magnitude (EVM) percentage circles, data striding, magnitude scaling, and phase offsets
- Plot format and style customization

The following topics describe the Vector Scope and how to use it:

- Vector Scope Window
- Complex Data Display Window
- Vector Scope Commands and Toolbars
- Starting Vector Scope
- Loading Data Sets
- Viewing Data Sets and Data Points

- [Saving Data Sets](#)
- [Printing Data Plots](#)
- [Customizing Plot Appearance](#)





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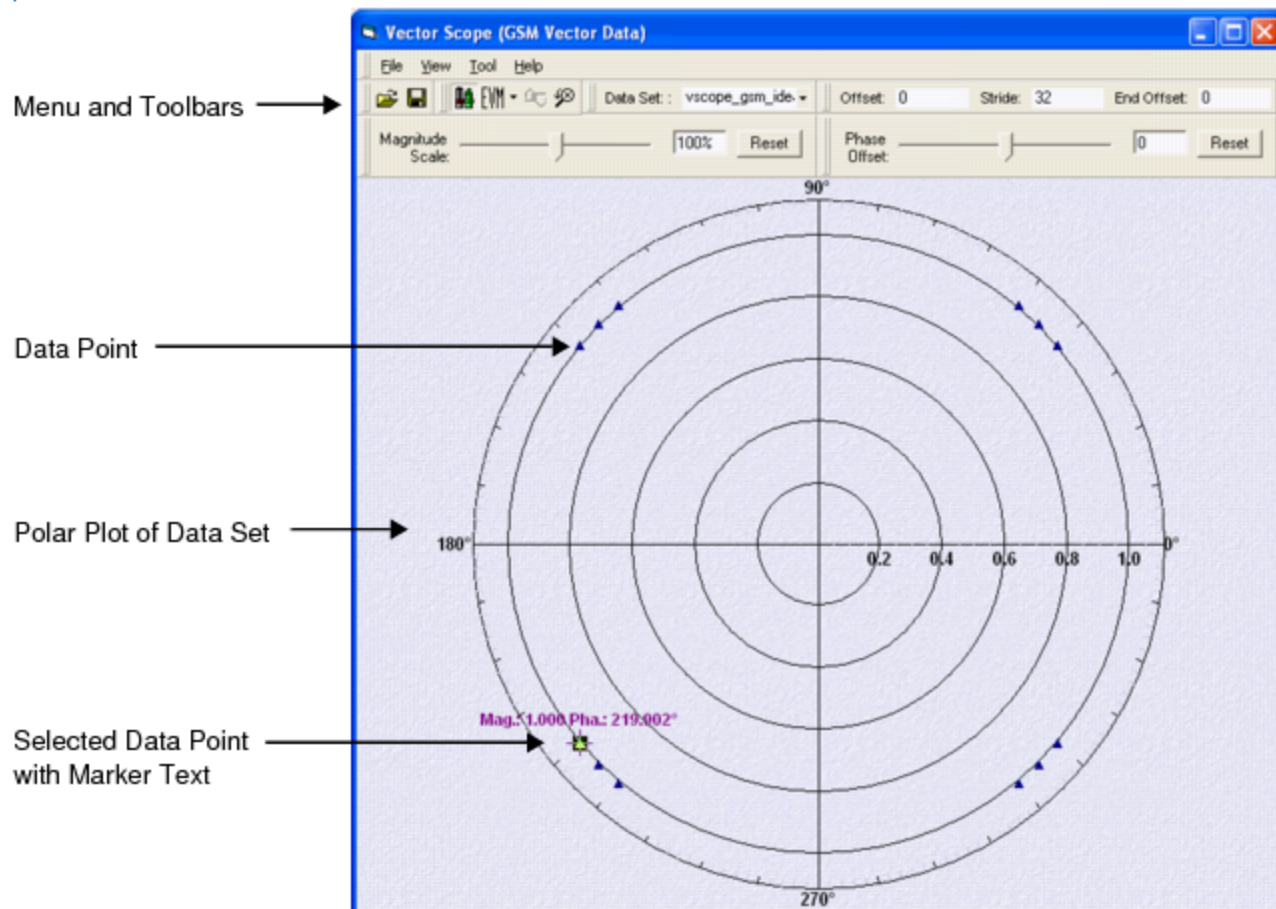
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Vector Scope : Vector Scope Window

Vector Scope Window

The figure shows the Vector Scope window with a number of complex data points displayed in a polar plot. Each data point is represented by a symbol (in this case a triangle) used for all data points in the data set.



Vector Scope Window

A data point in the lower left quadrant of the plot is shown selected, with markers turned on. With markers on, the arrow pointer changes to crosshairs and the polar coordinates of the data point are

displayed above the point. When a data point is selected, a black square appears behind the data point.

For more information, see [Vector Scope Commands and Toolbars](#).

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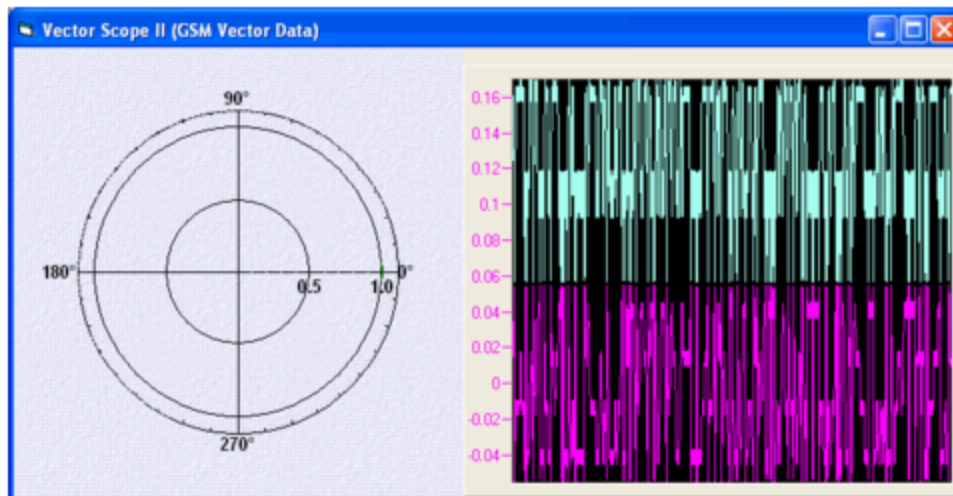


Vector Scope : Complex Data Display Window

Complex Data Display Window

You can start the Vector Scope from a test program executing in debug mode. For a DSPWave holding complex data, the DSPWave Plot method starts a complex data display window labeled Vector Scope II. As shown in the following figure, the display shows:

- A polar plot of the complex data set on the left
- Time plots (real and imaginary) of the complex data set on the right



Complex Data Display Window

The complex data display window shows compact views of the polar plot and the separate real and imaginary samples for a complex data set, useful for quick evaluation of program changes during debugging.

The window also provides easy access to both the Vector Scope and WaveScope displays:

- To open a Vector Scope, double-click anywhere on the polar plot.

A Vector Scope window appears with the data set loaded.

- To open two WaveScopes for the real and imaginary data samples, double-click anywhere on the time plots.

Two WaveScope windows open with real and imaginary data loaded.
For more information, see [About WaveScope](#).

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Vector Scope : Vector Scope Commands and Toolbars

Vector Scope Commands and Toolbars

The Vector Scope has a menu bar and several toolbars, some of which duplicate menu commands. This section includes the following topics:

- [Vector Scope Menus](#)
- [Vector Scope Toolbars](#)



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Vector Scope : [Vector Scope Commands and Toolbars](#) : Vector Scope Menus

Vector Scope Menus

The Menu toolbar provides the following menu selections for Vector Scope commands:

- [File Menu](#)
- [View Menu](#)
- [Tool Menu](#)
- [Help Menu](#)

File Menu

[File](#) menu commands are:

Open	 (Ctrl+O) Opens dialog to choose a data set file to display. File formats can be DSPWave (.txt) or IMAGE Wave (.wav). Vector Scope can load up to 10 data set files.
Save	 (Ctrl+S) Save selected data set to a file. Vector Scope saves only to .txt files. If the data set is loaded from a file, the file is updated if changes have been made. If the data set is displayed from a program, Save is equivalent to Save As.

	After a data set is saved, the data set name is updated and Vector Scope controls are reset.
Save As	Opens dialog to choose directory and file. See Save .
Close All	Closes all Vector Scope windows invoked by the VBT Plot method.

View Menu

[View](#) menu commands are:

Show/Hide Data Sets	Displays a list of all data sets loaded and controls whether data sets are shown or hidden: • If a data set is shown, a check mark appears before its name. • To show or hide a data set, select the data set name. • To show or hide all data sets, select the corresponding command. • If you select a hidden data set, modifications with the Magnitude Scale, Phase Offset, and Stride toolbars are still applied.
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Marker



Turns markers on or off. A marker is a text string near a data point showing the complex data values.

- When markers are turned on, the arrow pointer changes to crosshairs and data values are displayed when the arrow is over a data point or when a data point is selected. If you move the pointer off the data point, the data label persists only if the data point is selected.
- When you turn markers off, all marker data values are removed from the plot.

EVM	 Displays a circle of a specific error vector magnitude (EVM) allowing you to evaluate the error in a cluster of data points. EVM describes the modulation accuracy of a transmitted signal. The Vector Scope is optimized for modulation schemes that have a constant signal magnitudes, such as GMSK. The error circle scales in size along any radius from the origin.
Data Table	Displays information about the selected data point above the plot.
Legend	Displays a list of the symbols assigned to each shown data set to the right of the plot.
Overtrace	 Adds new data sets created during program execution without clearing previous data sets. Does not operate in stand-alone mode. See Using Overtrace.
Undo Zoom	 (Ctrl+Z) Zooms out, showing the complete plot. To zoom in, see Zooming In On a Plot Region.
XY Format	Plots data using rectangular coordinates and axes.

Tool Menu

Tool menu commands are:

Customize	Opens a dialog to allow custom settings for different aspects of a plot. See Customizing Plot Appearance .
Export	Opens a dialog to allow exporting a plot to a file, the clipboard, or a printer. See Saving Data Sets and Printing Data Plots .

Help Menu

Help menu commands are:

Vector Scope Help	(F1) Opens the Vector Scope topic in IG-XL Help.
About the Vector Scope	Opens a dialog with tool information including version.

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[Vector Scope](#) : [Vector Scope Commands and Toolbars](#) : Vector Scope Toolbars

Vector Scope Toolbars

The Vector Scope has a standard toolbar with buttons to open and save files and a View toolbar with a subset of the command available from the View menu.

In addition, Vector Scope has the following toolbars:

- [Data Set Toolbar](#)
- [Stride Toolbar](#)
- [Magnitude Scale Toolbar](#)
- [Phase Offset Toolbar](#)

For more information, see [Vector Scope Menus](#).

Data Set Toolbar

The Data Set toolbar provides a control for selecting a data set for specific operations such as offset and stride. The drop-down list shows all loaded data sets.



Data Set Toolbar

Stride Toolbar

The Stride toolbar provides controls for setting offsets and stride in a data set.



Stride Toolbar

The controls are:

Offset	Removes a number of samples starting from the beginning of the data set.
Stride	Retains only every Nth sample, where N is the stride value specified. In other words, skips N-1 samples between samples retained.
End Offset	Removes a number of samples starting from the end of the data set.

Magnitude Scale Toolbar

The Magnitude Scale toolbar provides controls for scaling all magnitudes in a selected data set.



Magnitude Scale Toolbar

The controls are:

Magnitude Scale (slider)	Scales the magnitudes of the points in the data set to values between 1% and 200% of the original value.
Magnitude Scale text box	Displays magnitude scale factor. This control also scales the magnitudes of points in the data set to values by percentage value typed in box. Value must be between 1% and 200%.
Reset	Resets magnitude scale to 100%.

Phase Offset Toolbar

The Phase Offset toolbar provides controls for a phase offset to all data points in a data set.



Phase Offset Toolbar

The controls are:

Phase Offset (slider)	Offsets the phases of the points in the data set by a value in the range from -180 to 180 degrees from the original values.
Phase Offset text box	Displays phase offset.

Reset

Offsets the phases of points in the data set by the value typed in the box. Value must be between -180 and 180.

Resets phase offset to 0 degrees.

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Vector Scope : Starting Vector Scope

Starting Vector Scope

You can use the Vector Scope in either stand-alone or integrated mode:

- In stand-alone mode, you can view, modify, and save data set files, but you cannot view data sets while debugging a program.
- In integrated mode, you can view waveform data in complex DSPWaves and modify and save data sets and plots to files.

The mode depends on how you start Vector Scope.

The ways to start Vector Scope are:

- Starting Vector Scope from a Test Program
- Starting Vector Scope from TDE
- Starting Vector Scope from the Command Prompt

Starting Vector Scope from a Test Program

If your program stores waveform data in a complex DSPWave, you can start the Vector Scope using the `Plot` method. DataTool must be in debug mode. Vector Scope starts in integrated mode.

If the waveform data is not complex, the `Plot` method starts WaveScope instead.

To start Vector Scope from a test program in debug mode:

1. Add the `Plot` method to the complex DSPWave you want to display.

2. Run the subroutine, function, or test program.

The complex data display window appears. See [Complex Data Display Window](#).

3. To start the full Vector Scope, double-click anywhere on the polar plot.

The Vector Scope window appears with the complex data set loaded, running in integrated mode:

- Each DSPWave starts a new Vector Scope.
- Vector Scope plots multisite DSPWaves for all sites on one display in the same color.
- If DSPWave data changes, and a Vector Scope is open, the open display is updated.
- If View>Overtrace is turned on, new data is appended to the existing data and displayed.

Vector Scope can load up to 10 data sets.

Starting Vector Scope from TDE

Vector Scope is available in TDE. When you start Vector Scope from TDE, it operates in stand-alone mode.

To start Vector Scope from TDE:

- Select TDE>Tools>VectorScope.

The Vector Scope window appears in TDE.

Starting Vector Scope from the Command Prompt

Vector Scope is available from the Windows command prompt. When you start Vector Scope from the command prompt, it operates in stand-alone mode.

To start Vector Scope from the command prompt:

1. Select Start>Programs>Accessories>Command Prompt.

The Command Prompt window appears.

2. Type vectorscope.

The Vector Scope window appears.

The Vector Scope executable is [vectorscope.exe](#) and is located in the ...\\Teradyne\\IG-XL\\V.xx.xx_yyy\\bin folder.

The [vectorscope](#) command accepts one or more data set files as arguments. After Vector Scope starts, it loads the named data sets. The maximum number of data sets is 10.

		
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Vector Scope : Loading Data Sets

Loading Data Sets

You can load one or more data sets (up to 10) into Vector Scope. Each data set is plotted with a different symbol and color by default. (To change symbol types and colors, see [Changing Data Point Type and Color](#).)

Vector Scope can open files in either the IG-XL DSPWave (.txt) format or IMAGE Wave (.wav) format. You load data sets in one of the following ways:

- [Loading Data Sets Using VBT in TDE](#)
- [Loading Data Sets Using the Open Command](#)
- [Loading Data Sets from the Command Window](#)
- [Loading Multiple Data Sets](#)

Loading Data Sets Using VBT in TDE

During program debug, the [Plot](#) method automatically loads complex waveform data held in a DSPWave into the complex data display window and Vector Scope. The Plot method accepts a string with a data set name. See [Starting Vector Scope from a Test Program](#).

If View>Overtrace is turned on, each new set of data is appended to the data set in the display. (See [Using Overtrace](#).)

Loading Data Sets Using the Open Command

In either stand-alone or integrated mode, you can load data sets using the [Open](#) command. To load a data set:

1. [Select](#) File>Open or toolbar button.

The Open Data File dialog appears.

2. [Navigate](#) to the folder and file you want to open.

The data points in the set appear on the plot and the name of the file appears in the Data Set toolbar.

Loading Data Sets from the Command Window

If you start Vector Scope in stand-alone mode from the Windows command prompt, Vector Scope can load one or more files after it starts.

To load one or more data sets:

- [List](#) one or more data set file names as arguments after the `vectorscope` command.

Vector Scope starts and then loads each file listed, up to 10 files.

For example, the following command starts Vector Scope and loads two data sets:

```
vectorscope gsm_ideal.txt gsm_capture.txt
```

Loading Multiple Data Sets

You can load more than one data set using Vector Scope in several ways:

- If you iterate through multiple sites in debug mode, Vector Scope loads data for each site as a data set, with all sets in the same color. See [Loading Data Sets Using VBT in TDE](#).
- If you [turn on](#) View>Overtrace, Vector Scope adds additional data sets for a DSPWave to the plot. This only works in integrated (debug) mode. See [Using Overtrace](#).
- If you start Vector Scope from the command prompt, you can specify more than one file as an argument. Each file is loaded as a separate data set. See [Starting Vector Scope from the Command Prompt](#).
- You can use the [Open](#) command more than once to open more than one data set file. See [Loading Data Sets Using the Open Command](#).

In all cases, Vector Scope can load a maximum of 10 data sets.

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Vector Scope : Viewing Data Sets and Data Points

Viewing Data Sets and Data Points

With one or more data sets loaded, you can use Vector Scope features to examine data sets and individual data points. You can also zoom in on a region and modify individual data sets. This section includes the following topics:

- | | |
|---|-----------------------------------|
| • | Displaying Data Sets |
| • | Zooming In On a Plot Region |
| • | Displaying Data Point Information |
| • | Modifying a Data Set |

You can also customize many display and style characteristics of a data plot. See Customizing Plot Appearance.

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Vector Scope : Viewing Data Sets and Data Points : Displaying Data Sets

Displaying Data Sets

Vector Scope displays one or more data sets on the charting area of the window. After the data sets are loaded, you control which ones are shown or hidden, the plot type (polar or XY), whether there is a legend of plot symbols, and whether overtrace is on (integrated debug mode only). See the following topics:

- Showing and Hiding Data Sets

- Displaying a Symbol Legend

- Using Overtrace

- Changing Plot Type

For more information, see [Viewing Data Sets and Data Points](#).

Showing and Hiding Data Sets

You can show or hide individual data sets in the display or show or hide all of them at once. To show or hide an individual data set:

1. [Select](#) View> Show/Hide Data Sets.

A menu appears with a list of all loaded data sets.

2. [Select](#) a data set name to show or hide it.

The names data set are shown or hidden on the display. The next time you select the menu, only the shown data set or sets will have a check mark.

Displaying a Symbol Legend

Vector Scope can display a list of the loaded data sets and the symbol assigned to each one. The list is located on the right side of the display window.

To display the legend:

- [Select View>Legend.](#)

The legend for all shown data sets appears on the right side of the display.

Using Overtrace

During test program debugging using the DSPWave.Plot method to load data sets, you can set Vector Scope to load a new data set to the same display each time new data is created.

To put Vector Scope in Overtrace mode:

- [Select View>Overtrace](#) or click the toolbar button.

Vector Scope will load a new data set on each Plot execution.

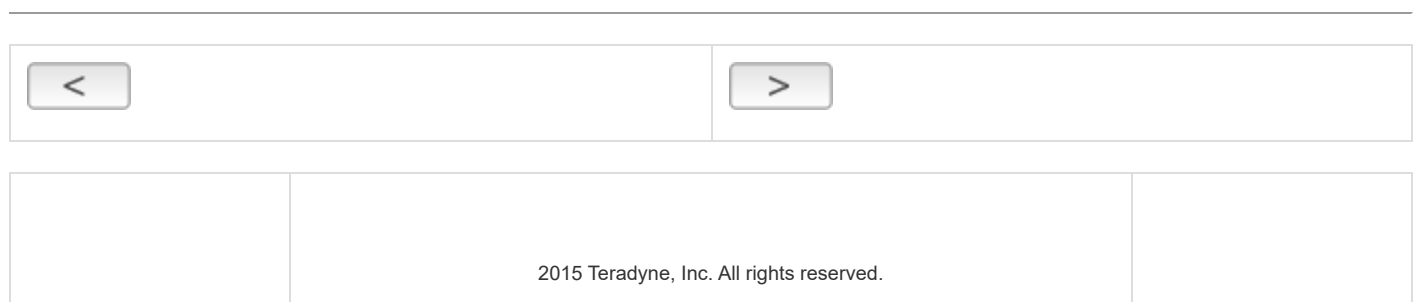
Changing Plot Type

Vector Scope can plot complex data using either polar coordinates (magnitude and phase) or XY coordinates (real and imaginary). Polar coordinates are the default.

To display complex data using XY coordinates:

- [Select View>XY Format.](#)

The chart area display shows XY axes and plots data by XY coordinates.



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Vector Scope : Viewing Data Sets and Data Points : Zooming In On a Plot Region

Zooming In On a Plot Region

Vector Scope supports one level of zoom. You can choose a rectangular region of the plot to be expanded (zoomed) to fill the entire chart area.

To zoom in on a region of the plot:

- Click in the plot near the data point or points you are interested in and drag the pointer to make a rectangle of an area to zoom.

The zoomed region expands to fill the chart area.

The zoomed region has a minimum size, which you must learn by trial and error.

To undo the zoom, do any of the following:

- Right click the chart area.

- Click the unzoom button.

- Select View>Undo Zoom.

For more information, see Viewing Data Sets and Data Points.



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Vector Scope : Viewing Data Sets and Data Points : Displaying Data Point Information

Displaying Data Point Information

Vector Scope provides several ways to investigate one or more individual data points, including a data table for selected data points, markers to place coordinates on the plot, and an EVM evaluation tool.

See the following topics:

- [Displaying a Data Table](#)
- [Using Markers](#)
- [Evaluating EVM](#)

For related topics, see [Viewing Data Sets and Data Points](#).

You can also customize display and style characteristics of the data plot. See [Customizing Plot Appearance](#).

Displaying a Data Table

Although it is hidden by default, Vector Scope can display a table of data for an individual data point. The table lists the data set, data size, and polar and XY data values.

To display a data table in the chart area:

- [Select View>Data Table.](#)

The data table appears at the top of the display.

To display the data for a data point:

- Click a data point to select it.

A block square background appears behind the data point, and the data appears in the table.

Using Markers

A marker is a text label near a data point. After you turn markers on, you can display markers in several ways.

To turn on markers:

- Select View>Marker or click the markers button.

Markers are turned on and display data under the pointer.

To see the coordinate values of a single data point:

- Hold the pointer over a data point.

The marker displays the data values for the data point while the pointer is over it.

To browse the coordinate values of several data points:

- Hold the pointer over each data point in turn.

The marker displays data values for each data point while the pointer is over the point.

To keep marker text on the plot after moving to another data point:

- Click to select the data point.

The marker stays after you move the pointer. You can accumulate any number of markers in this way.

To turn markers off:

- Select View>Marker or click the markers button.

Markers are turned off when the pointer is over a data point, and all current markers are removed from the display.

For more information, see Vector Scope Window.

Evaluating EVM

Error vector magnitude (EVM) describes the modulation accuracy of a transmitted signal. The Vector Scope displays a specific EVM value as a circle whose radius is a percentage of the ideal vector, allowing you to evaluate the error in a cluster of data points.

In a plot in XY Format, the circle is an ellipse if the coordinates have different scales.

The Vector Scope is optimized for modulation schemes that have constant signal magnitudes, such as GMSK. The error circle scales in size along any radius from the origin.

To evaluate the EVM of a set of data points:

1.

Select View>EVM>N% or the EVM button>N%, where N is a percentage of an ideal vector.

The pointer becomes a circle whose diameter changes with distance from the origin. A label displays the percentage.

2.

Move the EVM circle to a set of data.

If the data fits within the circle of percentage N, the EVM is less than N.

If the circle is much larger or smaller than the set of data points, you can adjust the size of the circle.

To turn off the EVM circle:

- Select View>EVM>Off or the EVM button>Off on the View toolbar.

The pointer returns to an arrow or crosshair.

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Vector Scope : Viewing Data Sets and Data Points : Modifying a Data Set

Modifying a Data Set

Using Vector Scope toolbars, you can modify a loaded data set. The plot adjusts to reflect the modifications you make.

To modify a loaded data set:

1. [Select a data set using the drop-down list on the Data Set toolbar.](#)

The name of the data set you selected appears in text box for the drop-down list.

2. [Change the data set using any combination of settings on the Stride, Magnitude Scale, and Phase Offset toolbars.](#)

The modifications are reflected in the plot of the selected data set.
The modifications are applied even if the data set is hidden. Also, if you save the selected data set, the original file is overwritten.
For more information, see [Viewing Data Sets and Data Points](#).



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Vector Scope : Saving Data Sets

Saving Data Sets

The Vector Scope can save modified data to a .txt file or a plot to an image file, in both stand-alone or integrated debug mode.

✓ Note:

If you save a data set file after you modify the data, the original data is overwritten. To save the original data, use the File>Save As command.

To save a data set to a new file:

1. Select a data set name from the drop-down list on the Data Set toolbar.

The data set name appears in the text box on the toolbar.

2. Select File>Save As.

The Save Data File dialog appears.

3. Select the file name and folder for the data set to be saved and click Save.

The file is saved and the dialog box closes.

You can also extract and save an image of a Vector Scope plot.

To save an image of a Vector Scope plot:

1. Select the Tool>Export command.

The Exporting dialog appears.

2.

Select the file type format from the Export section of the Export dialog. You can select MetaFile, bitmap (.bmp), JPEG (.jpg), or PNG (.png) formats.

3.

Select File as the Export Destination.

The Browse button becomes active.

4.

Click the Browse button.

The Save As dialog appears.

5.

Navigate to the folder where you want to save the file, specify the file name, and click Save.

The Save As dialog closes.

6.

Specify the desired dimensions of the graphical representation of the display in the Object Size block.

7.

Click the Export button.

The display plot is saved. You can open it using a graphics viewer/editor application.



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Vector Scope : Printing Data Plots

Printing Data Plots

The Vector Scope can send data plots to a printer using the Export tool.

To print a Vector Scope plot:

1.

Select Tool>Export.

The Exporting dialog appears.

2.

Select the MetaFile format from the Export block.

3.

Select Printer as the Export Destination.

4.

Specify the desired dimensions of the graphical representation of the display in the Object Size block.

5.

Click the Print button.

The Printing dialog appears.

6.

Specify the Printer destination and the Printing Style.

7.

Click OK.

The display plot is sent to the printer.

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Vector Scope : Customizing Plot Appearance

Customizing Plot Appearance

You can customize aspects of one or more data sets in a Vector Scope plot. Select **Tool>Customize** to open the Customization dialog.

The most common customizations are:

- Changing Data Set Plot Style
- Changing Data Point Type and Color

Changing Data Set Plot Style

The Vector Scope offers three visual styles for plotting a data set on the plot:

Points	Places an individual symbol at the exact data point on the Vector Scope.
Line	Connects a line between each data point.
Points+Line	Is a combination of Points and Line.

When you load and plot multiple data sets on the display, different point types and colors are used. To help distinguish between data sets, you can change the point and line styles. You can also change the symbols and colors of individual data sets.

To change data set plot style:

1. Select a data set from the drop-down list in the Data Set toolbar.
2. Select **Tool>Customize**.

The Customization dialog appears.

3. [Select the](#) Plot tab on the Customization dialog box.
4. [Select the desired data plot style](#) (Line, Points, or Points+Line) from the Plot Style box.
5. [Click](#) Apply.

The plot style is immediately applied to the plotted data set.

6. [Click](#) OK to complete the plot style change.

For more information, see [Changing Data Point Type and Color](#).

[Changing Data Point Type and Color](#)

You can change the appearance of the data points in different data sets to point types (symbols) and colors other than the default values Vector Scope uses.

To change the point type and color used for a data set:

1. [Select](#) Tool>Customize.

The Customization dialog appears.

2. [Select the](#) Style tab on the Customization dialog.

3. [Select the desired data set in the](#) Data Set List box.

The color palette, point type, and line controls show the current settings for the data set.

4. [Select the desired color from the color palette](#).

Changes are applied to the plot as you make them.

5. [Select the desired point type from](#) Point Type drop-down list.

Changes are applied to the plot as you make them.

6. [Select the desired line type from](#) Line Type drop-down list.

Changes are applied to the plot as you make them.

7.	Click OK.
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The dialog closes.

		
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